

Lab Report: Spatial and Hierarchical Analysis of U-District Dining

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Tools Used: Python (Colab), Yelp Fusion API, Observable HQ, QGIS

1. Project Overview

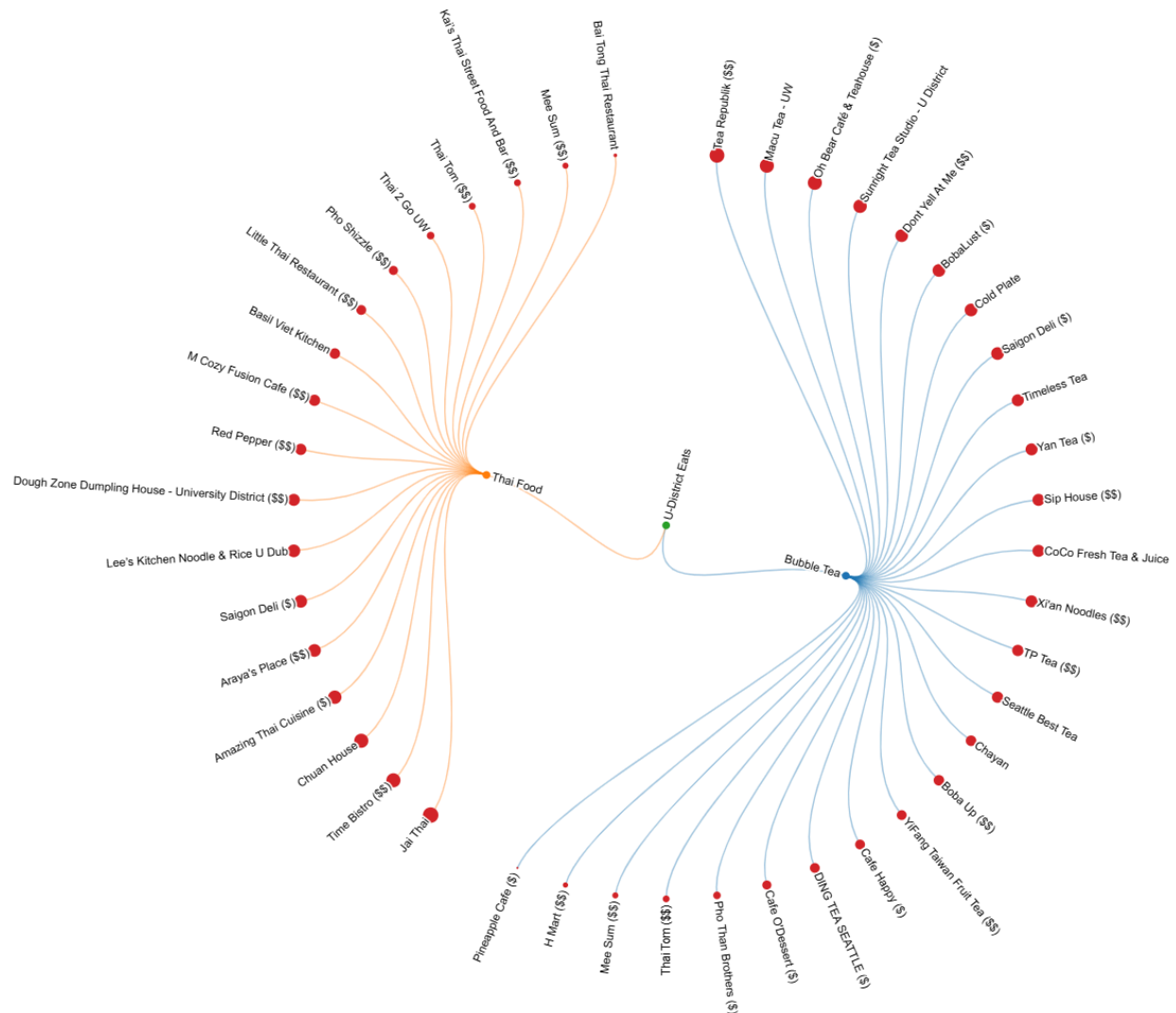
This project investigates the culinary landscape of the University District (U-District) in Seattle, specifically focusing on the distribution and pricing of **Thai Food** and **Bubble Tea**. By combining API data retrieval with spatial and hierarchical visualizations, we aim to understand how these two popular categories compete and cluster within the campus ecosystem.

2. Data Visualizations

A. Hierarchical Structure (Observable HQ)

The Radial Dendrogram illustrates the relationship between the parent category "U-District Eats," the sub-categories, and individual businesses with their respective price points.

- **Key Insight:** The tree shows a balanced variety in both categories. We observed some "cross-pollination" where certain venues (like Mee Sum) appear in multiple categories due to their diverse menus.

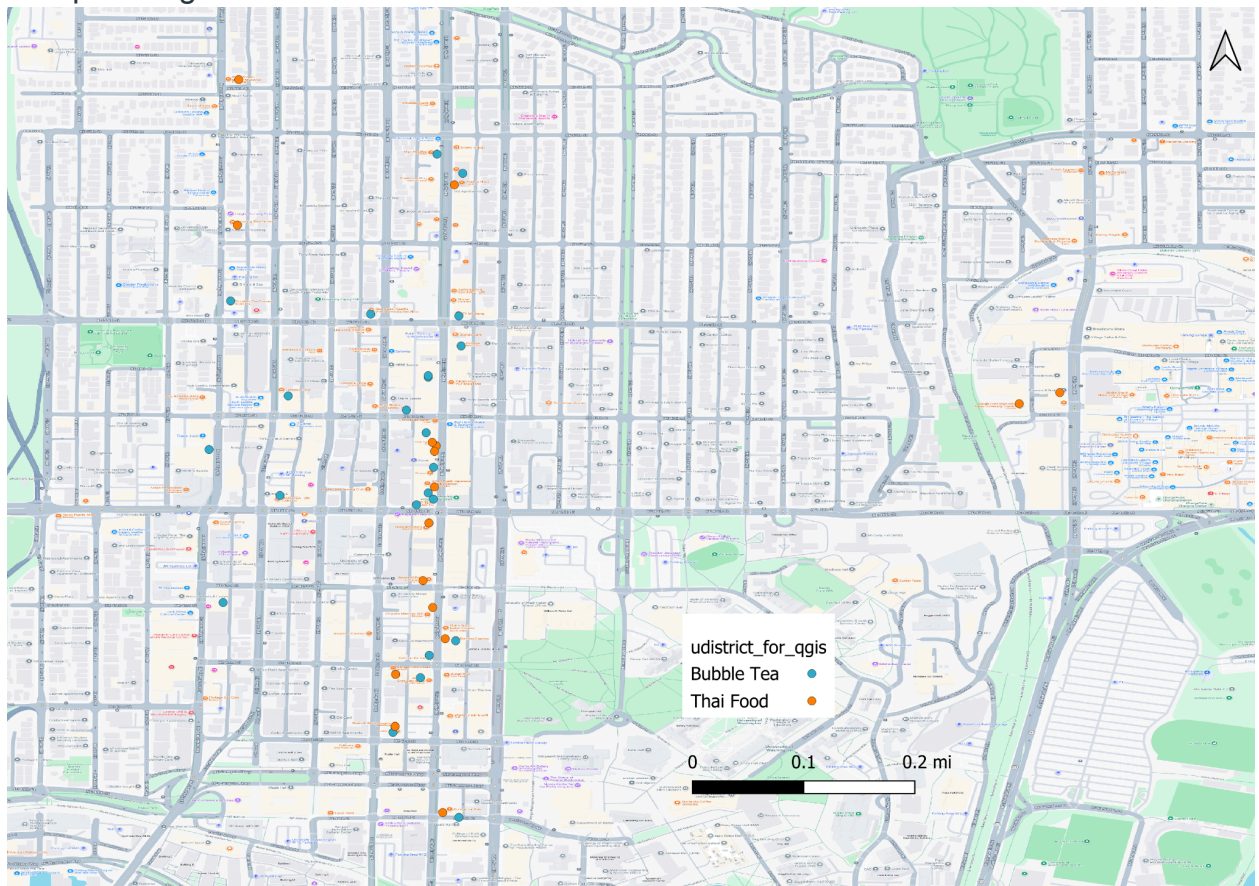


B. Spatial Distribution (QGIS Categorized Map)

Using QGIS and OpenStreetMap (OSM) data, we mapped the exact coordinates of each business.

- **Key Insight:** There is a clear "Bubble Tea Corridor" along **University Way NE (The Ave)**. Most Bubble Tea shops are tightly packed between 42nd and 45th St

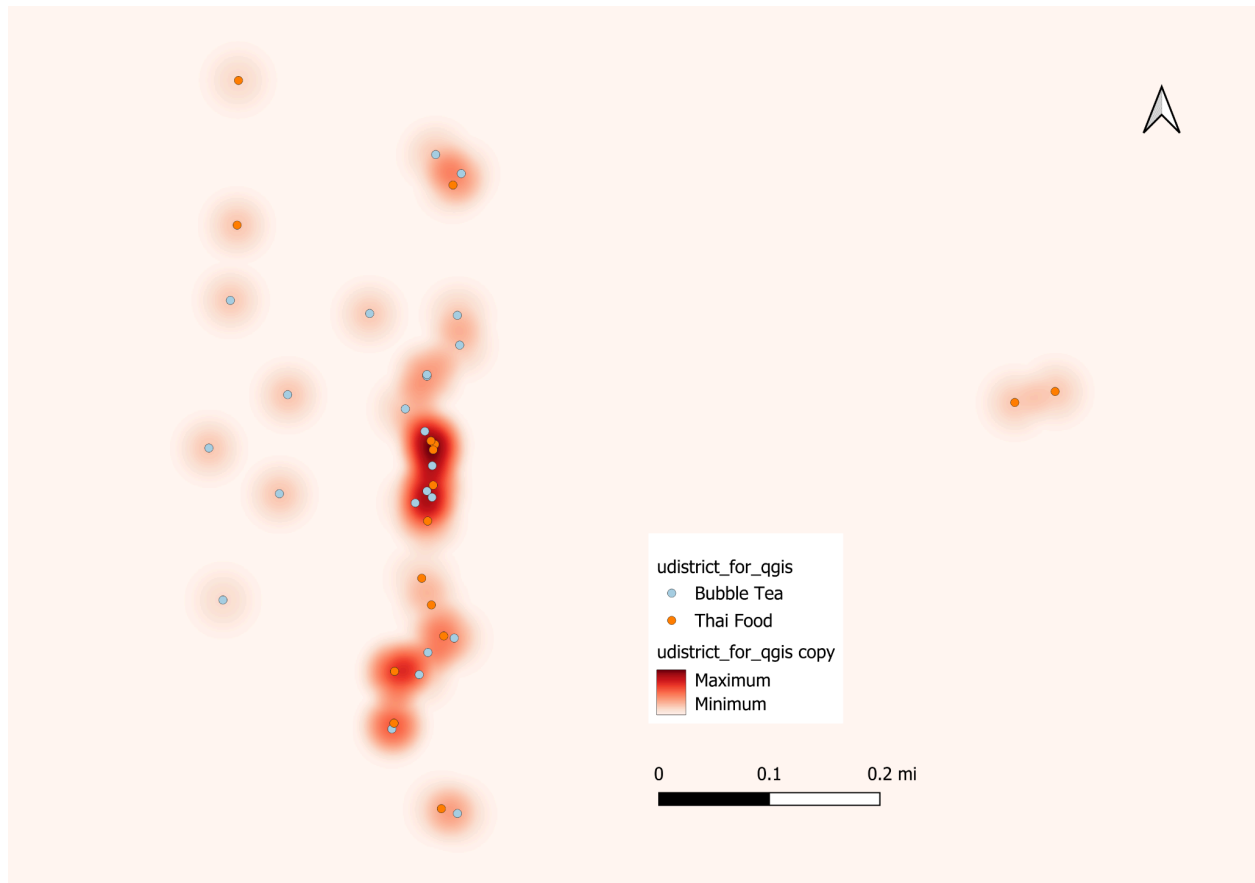
to capture high student foot traffic.



C. Density Analysis (QGIS Heatmap)

The heatmap identifies the "hotspots" of dining activity in the area.

- **Key Insight:** The highest density of dining options is concentrated on the southern end of The Ave, closest to the University of Washington campus entrance.



3. Technical Methodology

1. **Data Acquisition:** Data was scraped using the **Yelp Fusion API** via a Python script in Google Colab.
2. **Data Cleaning:** I converted price symbols (\$, \$\$) into numerical values and handled missing coordinate data to ensure compatibility with QGIS.
3. **Spatial Processing:** In QGIS, the `udistrict_for_qgis.csv` was imported as a Delimited Text Layer using **EPSG:4326 (WGS 84)** coordinates.
4. **Web Visualization:** The JSON-formatted data was imported into **Observable HQ** to generate a D3.js-based radial tree.

4. Final Conclusion

Our analysis reveals a strategic clustering of Bubble Tea shops in the high-traffic "core" of the U-District, likely targeting students during short breaks. Thai restaurants, while also present on The Ave, show a more "spread out" distribution, indicating they may serve as "destination" dining spots for the broader residential community beyond just the immediate student population.

Links for Review:

- **Google Colab Notebook:**
[<https://colab.research.google.com/drive/1jmr5RdxBCKQ1-TuxIVXhNU5-LQul5m mz?usp=sharing>]
- **Observable Notebook:** [<https://observablehq.com/d/7fd2a497087fa032>]